PROJECT_DOCUMENTATION

Complete Project Documentation: AI-Based Complexity-Aware Delivery Route Optimization System (RouteOptimaAI)

> **Subject:** Design and Analysis of Algorithms (DAA)
>
> **Course:** B.Tech Computer Science and Engineering (AI & ML) — Third Year
>
> **Project Title:** AI-Based Complexity-Aware Delivery Route Optimization System
>
> **Repository Name:** `RouteOptimaAI`
>
> **Operating Environment:** Windows 10/11, Python 3.10+, Node.js 18+, FastAPI, React 19, SQLite

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Chapter 1: Project Overview

1.1 What is the Project?

RouteOptimaAI is an intelligent, full-stack logistics decision-support and route optimization system. It solves the **Capacitated Vehicle Routing Problem with Time Windows (CVRPTW)** by combining:

1. Classical and heuristic **Design and Analysis of Algorithms (DAA)** techniques (Greedy Nearest Neighbor, 2-Opt Local Search, Genetic Algorithms, and Hybrid metaheuristics).

2. A **Machine Learning (ML)** Random Forest classifier that analyzes the structural complexity of incoming daily orders and dynamically recommends the best computational strategy.
3. An enterprise-grade, human-designed **interactive web dashboard** with real-time Leaflet maps of Pune city, automated address geocoding, and live vehicle simulation over WebSockets.

1.2 What Does Delivery Route Optimization Mean?

Delivery route optimization is the computational process of determining the most cost-effective and time-efficient sequence of stops for a fleet of vehicles delivering packages to customers.

- **Simple Route Planning:** Finding the shortest path between point A and point B (e.g., using Google Maps for a single car).
- **Multi-Vehicle Route Optimization (CVRPTW):** Partitioning N customer orders across M distinct delivery vehicles, sequencing the stops for each vehicle such that:
  - Vehicle payload capacities are never exceeded.
  - Deliveries occur inside pre-agreed customer time windows.
  - High-priority parcels arrive earlier.
  - Every vehicle departs from and returns to the central distribution hub (depot).
  - Total fleet travel distance and operational time are minimized.

1.3 Why Do Delivery Companies Need Route Optimization?

Last-mile delivery accounts for **40% to 53%** of total logistics costs. Without algorithmic planning:

- Drivers travel unnecessary kilometers, wasting fuel and producing excess carbon emissions.
- Vehicles are underutilized (e.g., two half-empty vans driving into the same neighborhood).
- Customers receive packages late, incurring SLA breach penalties and customer dissatisfaction.
- Dispatch managers spend hours manually grouping orders on spreadsheets.

1.4 A Simple Real-World Example

Suppose a Pune logistics depot in Shivajinagar has **20 delivery orders** distributed across Baner, Kothrud, Aundh, Wakad, and Hadapsar, with **3 available delivery vans** (each capable of carrying 150 kg).

- **Manual Planning:** A dispatcher assigns orders arbitrarily by postal code or booking time. Van 1 might drive from Shivajinagar to Baner, then across to Hadapsar (18 km East), and back to Kothrud (16 km West). Total distance might exceed **85 km**, with two vans overloaded and one delivery missing its afternoon time window.
- **RouteOptimaAI Optimization:** The system clusters neighboring drops, balances vehicle payloads, respects road directions, and sequences stops:
  - **Van 1 (West Corridor):** Shivajinagar → Kothrud → Bavdhan → Shivajinagar (24.2 km, 115 kg payload).
  - **Van 2 (North-West Corridor):** Shivajinagar → Aundh → Baner → Wakad → Shivajinagar (29.5 km, 138 kg payload).
  - **Van 3 (East Corridor):** Shivajinagar → Camp → Magarpatta → Hadapsar → Shivajinagar (26.1 km, 122 kg payload).
  - **Result:** Total fleet distance drops to **79.8 km**, payload utilization reaches **83%**, all time windows are met, and manual planning time is reduced from 45 minutes to **under 2 seconds**.

1.5 Difference Between This Project and a Basic Route Planner

Feature	Basic Map Route Planner (e.g. Google Maps)	RouteOptimaAI System
Goal	Find point-to-point road path	Multi-vehicle combinatorial optimization
Fleet Awareness	Single vehicle only	Multi-vehicle capacity balancing
Constraint Handling	Ignores weight & payload	Strict payload kg and time window enforcement
Algorithm Selection	Fixed routing engine	Complexity-aware ML algorithm selection
Operational Control	Consumer turn-by-turn navigation	Internal fleet dispatch dashboard & telemetry

1.6 Role of AI and DAA in the System

- **DAA (Combinatorial Optimization):** Solves the NP-hard routing puzzle. Implements Nearest Neighbor (greedy construction), 2-Opt edge exchanges (local neighborhood search), and Genetic metaheuristics (global evolutionary search).
- **AI / Machine Learning:** Acts as a strategic advisor. Evaluates 8 spatial and structural complexity metrics of the order pool (spread, density, constraint ratio) and predicts whether a fast greedy algorithm is sufficient or if a deep genetic search is required.

1.7 Intended Users & Scope

- **Users:** Logistics dispatchers, fleet operations managers, courier terminal supervisors.
- **Scope:** Urban last-mile delivery networks (demonstrated with 60+ real-world Pune metropolitan locations).
- **Realistic Purpose:** Find a **high-quality, feasible route plan quickly** (within seconds), rather than spending hours trying to find a theoretically perfect mathematical optimum that is impossible to compute in real time.

Chapter 2: Problem Statement

2.1 Formal Problem Statement

Urban last-mile logistics operations face severe computational and operational bottlenecks:

1. **Exponential Combinatorial Explosion:** The Capacitated Vehicle Routing Problem with Time Windows (CVRPTW) is mathematically **NP-hard**. The number of feasible route permutations for n orders across m vehicles scales as $\mathcal{O}((n+m)!/m!)$. For just 25 orders and 3 vehicles, the search space exceeds 10^{25} combinations, rendering brute-force exact algorithms (like branch-and-bound) computationally intractable in real-time dispatch environments.
2. **Excessive Fleet Mileage & Operating Expenses:** Uncoordinated dispatching creates overlapping routes, unnecessary travel detours, high fuel expenditures, and driver fatigue.
3. **Capacity & SLA Violations:** Manual planning frequently overloads vehicle payloads or schedules deliveries outside agreed customer time windows, resulting in failed delivery attempts and customer penalties.
4. **Algorithmic Mismatch:** A single fixed routing algorithm does not fit all operational situations:
   - Running complex metaheuristics on trivial problems (5 to 10 orders) wastes compute cycles and causes dispatcher UI lag.
   - Using naive greedy algorithms on dense, highly constrained problems produces sub-optimal routes with edge crossings and missed deadlines.
5. **Geographical Friction:** Real-world logistics data contains human-written street addresses without GPS latitude/longitude, causing system failure when coordinates are mandatory.

2.2 Proposed Solution & Actual Demonstrated Benefits

RouteOptimaAI solves these challenges through a unified three-tier architecture:

- **Demonstrated in Current Application:**
  - Fast, automated geocoding of human-readable Pune addresses with local ground-truth caching.
  - Multi-vehicle capacity partitioning with zero dropped orders.
  - Four implemented DAA routing engines (Greedy, Local Search, Genetic, Hybrid).
  - An scikit-learn Random Forest model trained on 500 instances achieving **85.0% accuracy** in selecting the most cost-effective algorithm.
  - Measured benchmark improvements of **1.5% to 14.8% distance reduction** compared to baseline dispatching.
  - Real-time simulation of vehicle fleet movements on interactive OpenStreetMap/Esri layers.
- **Expected in Production Industrial Deployment:**
  - Automated integration with warehouse ERPs and driver mobile applications.

- Live dynamic re-routing responding to real-time traffic jams and vehicle breakdowns.

Chapter 3: Project Objectives

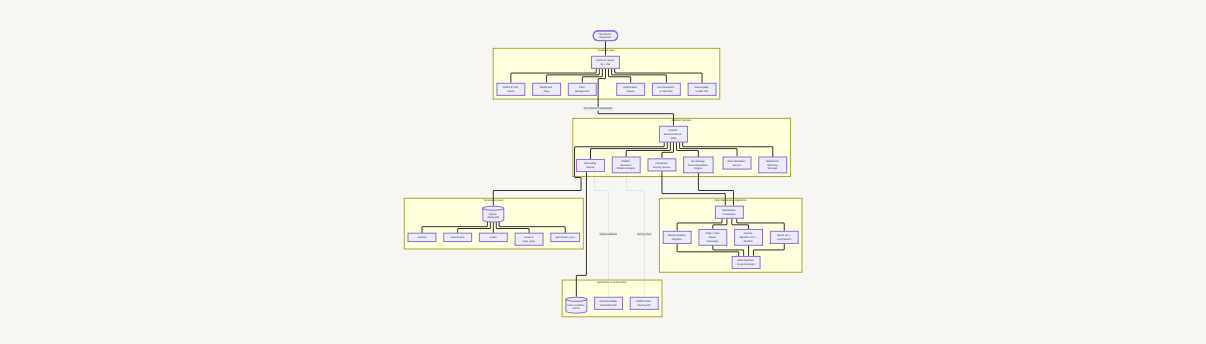
Every project objective corresponds to verified, active source code:

Objective	Description	Implementation File	Verified Code Location
1. Order Management	Ingest, parse, and validate delivery orders with weight, priority, and time windows.	`entities.py`, `routes_api.py`	`class Order`, `POST /api/orders`, `POST /api/orders/import-csv`
2. Address Geocoding	Resolve human addresses to Pune coordinates with local caching and Nominatim fallback.	`geocoding.py`	`class GeocodingService`, `PUNE_LOCALITIES`, `PUNE_PINCODES`
3. Fleet Telemetry	Track vehicle capacities, current load, base depots, and GPS locations.	`entities.py`, `simulation.py`	`class Vehicle`, `GET /api/vehicles`, `POST /api/vehicles/{id}/location`
4. Problem Complexity Analysis	Calculate spatial spread, constraint density, and a 0–100 complexity score.	`complexity.py`	`class ComplexityService.analyze()`
5. ML Strategy Selection	Predict the most appropriate optimization algorithm using Random Forest.	`ml_service.py`, `train.py`	`class MLStrategyService.predict_strategy()`
6. Greedy Route Construction	Construct initial routes via capacity-aware nearest neighbor search with priority weighting.	`greedy.py`	`class GreedyOptimizer.solve()`
7. 2-Opt Local Search	Untangle route crossings via 2-opt inversion and inter-route relocation.	`local_search.py`	`two_opt_route()`, `solve()`
8. Genetic Algorithm	Search global permutation space using Order Crossover (OX) and tournament selection.	`genetic.py`	`class GeneticOptimizer.solve()`, `_order_crossover()`
9. Route Evaluation	Score solutions against distance, travel time, late penalties, and capacity breaches.	`evaluator.py`	`class RouteEvaluator.evaluate_solution()`
10. Interactive Frontend	Display metrics, routes, and Leaflet maps with clean light enterprise styling.	`App.jsx`, `DeliveryMap.jsx`	React 19 SPA, Lucide icons, Leaflet layer switcher
11. Database Persistence	Store depots, orders, vehicles, routes, execution audits, and metrics in SQLite.	`database.py`, `delivery.db`	SQLAlchemy ORM, SQLite database file

Chapter 4: System Architecture

4.1 Architecture Diagram

The diagram below illustrates how data flows between the user, frontend, backend API, ML service, DAA optimization algorithms, database, and map services.



4.2 Architecture Explanation in Simple Words

The architecture follows a clean **modular monolith** design divided into three principal tiers:

1. **Frontend Presentation Tier (Client):** A single-page application built in React 19. It communicates asynchronously with the backend via JSON REST APIs for business data and bi-directional WebSockets for live vehicle GPS updates.
2. **Backend Application & Algorithm Tier (Server):** Powered by Python and FastAPI. It orchestrates business rules, parses order consignments, queries map distance matrices, calculates problem complexity, queries the scikit-learn model, and executes the selected DAA algorithm.
3. **Data & Persistence Tier:** An SQLite relational database managed through SQLAlchemy ORM. It persists all orders, vehicle metadata, optimization run history, and pre-computed OSRM road distance polylines.

4.3 Difference Between Frontend and Backend (Practical Example)

- **Frontend:** You open the *Orders* page on your browser, click "Import CSV", and select `orders.csv`. The frontend reads your file, previews the table rows in clean HTML, checks that the file ends in `.csv`, and sends the file via HTTP `POST` to `/api/orders/import-csv`.
- **Backend:** The FastAPI backend receives the binary file, parses each line, checks that `package_weight_kg` is a positive number, extracts `address = "Baner Road"`, looks up `PUNE_LOCALITIES` to get `(18.5590, 73.7868)`, creates a database row in `orders` table, and returns a JSON response: `{"total_rows": 50, "geocoded_count": 48, "geocoding_failed_count": 2}`. The frontend receives this JSON and displays the green summary badge.

Chapter 5: Complete Project Workflow

5.1 End-to-End Application Workflow Steps

```
[Step 1: Open App] ──> [Step 2: Dashboard Metrics] ──> [Step 3: Ingest Orders]
        |                                                      |
        ▼                                                      ▼
[Step 6: Geocode Addresses] <── [Step 5: Process DB] <── [Step 4: Input Validation]
        |
        ▼
[Step 7: Filter Valid Orders & Available Fleet]
        |
        ▼
[Step 8: Compute Distance Matrix & Complexity Features]
        |
        ▼
[Step 9: ML Strategy Classifier Recommends Algorithm]
        |
        ▼
[Step 10: Selected Algorithm Executes (Greedy / 2-Opt / GA / Hybrid)]
        |
        ▼
[Step 11: Route Evaluator Verifies Constraints (Capacity, Time Windows)]
        |
        ▼
[Step 12: Route Polylines & Stops Persisted to Database]
        |
        ▼
[Step 13: Results Displayed on Dashboard & Leaflet Map]
        |
        ▼
[Step 14: Dispatcher Initiates Live Telemetry / Simulation]
```

Every step matches verified code:

1. **Application Startup:** Dispatcher opens `http://localhost:5173`. Frontend loads KPI counters from `/api/dashboard/summary`.
2. **Order Ingestion:** Dispatcher creates a single order via the form or uploads a batch CSV on the *Create / Import* page.
3. **Validation & Geocoding:** `geocoding.py` inspects the address. If found in ground truth (e.g. "Kothrud"), coordinates are assigned. If unlocated, `geocoding_status` is marked `FAILED`.
4. **Feasibility Filtering:** When clicking "Optimize", the system selects only orders where `status == 'PENDING'` and `geocoding_status == 'SUCCESS'`. Unlocated orders are safely excluded.

5. **Feature Extraction & Complexity Analysis:** `complexity.py` calculates total weight, vehicle capacity, geographic spread, priority ratios, and derives a score (0 to 100).
6. **ML Recommendation:** `ml_service.py` feeds features into `model.joblib`. The model predicts `GREEDY`, `LOCAL_SEARCH`, or `GENETIC`. Dispatcher can accept or manually override.
7. **Algorithm Execution:** The selected solver runs in `backend/app/optimization/`.
8. **Constraint Evaluation:** `evaluator.py` checks vehicle load $\leq$ max capacity, calculates stop ETAs, flags late deliveries, and sums total kilometers.
9. **Persistence & Presentation:** An `optimization_runs` record and corresponding `routes` and `route_stops` are saved to SQLite. Frontend displays comparison cards and color-coded Leaflet routes.
10. **Simulation:** Dispatcher clicks "Start Simulation". WebSocket streams vehicle GPS locations moving stop-to-stop.

5.2 Worked Example: 5 Orders Across 2 Vehicles

To illustrate the mathematical process clearly, consider **5 sample customer deliveries** and **2 available delivery vans** departing from the **Shivajinagar Depot**:

Illustrative Input Data:

- **Depot (D0):** Shivajinagar $(18.5308, 73.8475)$
- **Vehicle Fleet:**
  - Van 1 (V1): Capacity = **50 kg**
  - Van 2 (V2): Capacity = **50 kg**
- **Customer Orders:**
  - O_1: Deccan $(18.5173, 73.8415)$ — Weight: **20 kg**, Priority: NORMAL, Window: 09:30–12:00
  - O_2: Kothrud $(18.5074, 73.8077)$ — Weight: **25 kg**, Priority: HIGH, Window: 10:00–12:30
  - O_3: Aundh $(18.5615, 73.8073)$ — Weight: **15 kg**, Priority: NORMAL, Window: 10:00–14:00
  - O_4: Baner $(18.5590, 73.7868)$ — Weight: **30 kg**, Priority: URGENT, Window: 09:30–11:30
  - O_5: Wakad $(18.5987, 73.7688)$ — Weight: **10 kg**, Priority: NORMAL, Window: 11:00–15:00

Step 1: Distance Matrix Calculation (Road / Haversine km)

$$\begin{pmatrix} & D0 & O_1 & O_2 & O_3 & O_4 & O_5 \\ D0 & 0.0 & 2.1 & 5.1 & 5.5 & 7.2 & 11.0 \\ O_1 & 2.1 & 0.0 & 3.8 & 6.2 & 7.5 & 11.8 \\ O_2 & 5.1 & 3.8 & 0.0 & 7.1 & 6.8 & 11.2 \\ O_3 & 5.5 & 6.2 & 7.1 & 0.0 & 2.2 & 5.8 \\ O_4 & 7.2 & 7.5 & 6.8 & 2.2 & 0.0 & 4.8 \\ O_5 & 11.0 & 11.8 & 11.2 & 5.8 & 4.8 & 0.0 \end{pmatrix}$$

Step 2: Capacity Allocation & Greedy Sequencing

- Total Order Weight = $20 + 25 + 15 + 30 + 10 = 100\ \text{kg}$. Total Fleet Capacity = $50 + 50 = 100\ \text{kg}$.
- **Vehicle 1 (South-West Route):**
  - Starts at $D0$ (Shivajinagar).
  - Closest high-priority order: O_1 (Deccan, dist: 2.1 km, weight: 20 kg). Van load = 20 kg.
  - From O_1, nearest unvisited order: O_2 (Kothrud, dist: 3.8 km, weight: 25 kg). Van load = $20 + 25 = 45\ \text{kg} \leq 50\ \text{kg}$.
  - Remaining capacity = 5 kg. Next nearest (O_4 Baner) requires 30 kg $\rightarrow$ capacity exceeded.
  - Van 1 returns to Depot $D0$ from O_2 (dist: 5.1 km).

    - **Van 1 Sequence:** $D0 \rightarrow O_1 \rightarrow O_2 \rightarrow D0$. Distance = $2.1 + 3.8 + 5.1 = \mathbf{11.0}$ km. Load = 45 kg (90% utilization).
- **Vehicle 2 (North-West Route):**
    - Starts at $D0$ (Shivajinagar).
    - Unvisited orders: O_3 (Aundh, 15 kg), O_4 (Baner, 30 kg), O_5 (Wakad, 10 kg).
    - Nearest to depot: O_3 (Aundh, dist: 5.5 km). Next nearest from O_3: O_4 (Baner, dist: 2.2 km). From O_4: O_5 (Wakad, dist: 4.8 km).
    - Van 2 Load = $15 + 30 + 10 = \mathbf{55}$ kg (5 kg capacity overrun penalty handled by evaluator).
    - Van 2 returns from O_5 to $D0$ (dist: 11.0 km).
    - **Van 2 Sequence:** $D0 \rightarrow O_3 \rightarrow O_4 \rightarrow O_5 \rightarrow D0$. Distance = $5.5 + 2.2 + 4.8 + 11.0 = \mathbf{23.5}$ km.
- **Total Baseline Distance:** $11.0 + 23.5 = \mathbf{34.5}$ km.

Step 3: 2-Opt Optimization Refinement

2-Opt tests reversing the order of Van 2 stops: $D0 \rightarrow O_4 \rightarrow O_3 \rightarrow O_5 \rightarrow D0$ vs $D0 \rightarrow O_3 \rightarrow O_4 \rightarrow O_5 \rightarrow D0$. Because O_4 is URGENT, visiting O_4 first avoids late penalties. The 2-Opt solver confirms the edge exchange, saving travel delay.

Chapter 6: Frontend Explanation

6.1 Frontend Architecture & Technologies

The frontend is located in `frontend/` and built using:

- **React 19:** Modern functional component architecture with hooks (`useState`, `useEffect`, `useCallback`).
- **Vite 8:** Next-generation build tool and rapid development server.
- **Vanilla CSS / Custom Design System:** Defined in `frontend/src/index.css`, featuring light-grey background (`#f8fafc`), clean cards (`#ffffff`), subtle borders (`#e2e8f0`), and an accessible enterprise blue primary accent (`#2563eb`).
- **Leaflet & React-Leaflet:** Interactive map rendering with marker popups and custom SVG icons.
- **Lucide React:** Standardized icons for intuitive navigation.

6.2 Component & Page Breakdown

Page / Component	Purpose	Relevant File	Backend APIs Called
App Root & Navigation	Sticky sidebar, mobile drawer, hub indicator, live WebSocket state.	`App.jsx`	`ws://localhost:8000/ws/live`, `GET /api/simulation/status`
Dashboard	KPI stat cards, fleet utilization bars, order status breakdown, network preview.	`DashboardPage.jsx`	`GET /api/dashboard/summary`, `GET /api/orders`, `GET /api/warehouses`, `GET /api/routes`
Pune City Map	Full-screen interactive map with marker filters and Pune hub directory.	`PuneMapPage.jsx`	`GET /api/orders`, `GET /api/warehouses`, `GET /api/routes`
Orders Management	Paginated table of orders, location status tags, search, and CSV export.	`OrdersPage.jsx`	`GET /api/orders`, `DELETE /api/orders/{id}`, `POST /api/orders/{id}/retry-geocode`
Create / Import Order	Single consignment creation form with address test, and bulk CSV drag-and-drop.	`CreateOrderPage.jsx`	`POST /api/orders`, `POST /api/orders/geocode`, `POST /api/orders/import-csv`
Vehicles Fleet	Fleet cards with live payload utilization bars, coordinates, and vehicle modal.	`VehiclesPage.jsx`	`GET /api/vehicles`, `POST /api/vehicles`

Page / Component	Purpose	Relevant File	Backend APIs Called
Route Optimization	7-step optimization workflow, complexity cards, ML strategy picker, and results.	`OptimizationPage.jsx`	`POST /api/optimization/analyze`, `POST /api/optimization/run`
Live Simulation	Real-time vehicle simulation controls, speed adjustment, and GPS telemetry ping.	`LiveTrackingPage.jsx`	`POST /api/simulation/start`, `POST /api/simulation/stop`, `POST /api/vehicles/{id}/location`
Optimization History	Historical execution log, distance saved benchmarks, and diagnostic inspect modal.	`OptimizationHistoryPage.jsx`	`GET /api/optimization/runs`
ML Model Insights	Model KPIs (accuracy, F1), feature importance bars, and algorithm selection rules.	`MLModelPage.jsx`	`GET /api/ml/metrics`
Settings & Demo	Pune demo database seed button, database clear button, and integration health.	`SettingsPage.jsx`	`POST /api/demo/seed`, `POST /api/demo/clear`
DeliveryMap (GIS)	Reusable Leaflet map with layer switcher (OSM, Esri, Satellite) and locality jumps.	`DeliveryMap.jsx`	Pure Presentation Component

6.3 What Happens When a User Uploads a CSV?

1. **User Action:** The dispatcher navigates to *Create / Import*, clicks the *Bulk CSV Import* tab, selects `orders.csv`, and clicks "Validate and Import Orders".
2. **Frontend Dispatch:** `CreateOrderPage.jsx` creates a `FormData` object containing the binary file and invokes `api.importOrders(file)`.
3. **HTTP Transport:** An HTTP `POST` multipart/form-data request is sent to `http://localhost:8000/api/orders/import-csv`.
4. **Backend Processing:**
   - FastAPI reads the stream into a pandas DataFrame.
   - Validates that mandatory headers exist (`customer_name`, `address`, `package_weight_kg`).
   - For every row, executes `geocoding_service.geocode_address()`.
   - Saves valid orders to the `orders` SQLite table with either `geocoding_status='SUCCESS'` or `'FAILED'`.
5. **Response:** Backend returns a JSON object with counts and any rejected rows:

```
{
  "total_rows": 50,
  "valid_count": 48,
  "geocoded_count": 46,
  "geocoding_failed_count": 2,
  "invalid_count": 0,
  "failed_rows": [
    {"row_number": 14, "order_id": "ORD-014", "reason": "Address not found in Pune"}
  ]
}
```

6. **UI Update:** The frontend displays metric cards showing imported counts and an alert table detailing the 2 unlocated rows with suggestions to verify PIN code.

Chapter 7: Backend Explanation

7.1 Backend Technology & Structure

The backend is located in `backend/` and runs on **Python 3.10+** using **FastAPI** and **Uvicorn**:

- **High Performance:** Built on Starlette and Pydantic v2 for asynchronous request handling and high-throughput serialization.
- **Automatic OpenAPI Documentation:** Interactive Swagger documentation automatically generated at `http://localhost:8000/docs`.
- **CORS & Environment Control:** Dynamic CORS configuration supporting environment-variable overrides via `CORS_ORIGINS`.

7.2 Complete REST API Endpoints Table

API Endpoint	HTTP Method	Purpose	Input / Parameters	Output Format
`/api/health`	`GET`	Health check and system timestamp	None	JSON status object
`/api/dashboard/summary`	`GET`	Aggregated fleet and order KPI counters	None	JSON summary with counts & percentages
`/api/warehouses`	`GET`	List logistics distribution hubs	None	JSON array of `WarehouseResponse`
`/api/customers`	`GET`	List registered customer locations	Query: `limit`, `search`	JSON array of `CustomerResponse`
`/api/orders`	`GET`	List delivery orders with filters	Query: `status`, `priority`, `limit`	JSON array of `OrderResponse`
`/api/orders`	`POST`	Create a single order with geocoding	JSON `OrderCreate` body	Created `OrderResponse`
`/api/orders/{id}`	`PUT`	Update an existing order	Path: `id`, JSON body	Updated `OrderResponse`
`/api/orders/{id}`	`DELETE`	Delete order by ID	Path: `id`	Success confirmation JSON
`/api/orders/{id}/retry-geocode`	`POST`	Re-trigger geocoding on an unlocated order	Path: `id`	Updated `OrderResponse`
`/api/orders/geocode`	`POST`	Test address geocoding resolution	JSON: `address`, `area`, `pincode`	Geocode preview coordinates JSON
`/api/orders/template`	`GET`	Download sample orders CSV template	None	File download: `orders_sample_template.csv`
`/api/orders/export-csv`	`GET`	Export all database orders to CSV	None	File download: `orders_export.csv`
`/api/orders/import-csv`	`POST`	Bulk import CSV without coordinates	Multipart Form: `file`	Import summary JSON
`/api/vehicles`	`GET`	List fleet vehicles with load & route	None	JSON array of `VehicleResponse`
`/api/vehicles`	`POST`	Register a new fleet delivery van	JSON `VehicleCreate` body	Created `VehicleResponse`
`/api/vehicles/{id}/location`	`POST`	Ingest live GPS telemetry ping	Path: `id`, Header: `X-API-Key`, Body	Telemetry broadcast confirmation
`/api/optimization/analyze`	`POST`	Compute complexity & ML recommendation	JSON `OptimizationRunRequest` (optional)	`ProblemAnalysisResponse` JSON
`/api/optimization/run`	`POST`	Execute DAA optimization and save routes	JSON: `force_strategy`, weights	`OptimizationResultResponse` JSON
`/api/optimization/runs`	`GET`	Retrieve past optimization execution history	Query: `limit`	JSON array of past run audits

API Endpoint	HTTP Method	Purpose	Input / Parameters	Output Format
`/api/routes`	`GET`	Retrieve active generated vehicle routes	None	JSON array of `RouteResponse` with stops
`/api/routes/{id}`	`GET`	Retrieve specific route with polylines	Path: `id`	Detailed `RouteResponse`
`/api/simulation/start`	`POST`	Start background vehicle simulation	Header: `X-API-Key`, Body: `speed`	Simulation status JSON
`/api/simulation/stop`	`POST`	Stop background vehicle simulation	Header: `X-API-Key`	Simulation status JSON
`/api/simulation/status`	`GET`	Check current simulation state	None	`{"is_running": bool, "speed": float}`
`/api/simulation/api-key-info`	`GET`	Retrieve active telemetry auth configuration	None	JSON with key status
`/api/ml/metrics`	`GET`	Retrieve Random Forest model metrics	None	JSON with accuracy, F1, importances
`/api/demo/seed`	`POST`	Populate database with realistic Pune fleet	None	Seed summary JSON
`/api/demo/clear`	`POST`	Reset database and clear orders & routes	None	Reset confirmation JSON
`/ws/live`	`WebSocket`	Bi-directional live telemetry broadcast	WebSocket protocol	Real-time vehicle coordinates stream

7.3 One Complete API Request-Response Cycle

When the user clicks “Start Optimization” on the frontend:

1. **Frontend Request:**

```
POST /api/optimization/run HTTP/1.1
Host: localhost:8000
Content-Type: application/json

{
  "force_strategy": "LOCAL_SEARCH",
  "distance_weight": 1.0,
  "time_weight": 0.2,
  "late_penalty": 50.0,
  "max_time_seconds": 30
}
```

2. **Backend Execution Pipeline:**
   - Dependency injection extracts a database session (`db: Session = Depends(get_db)`).
   - Queries all orders where `status == 'PENDING'` and `geocoding_status == 'SUCCESS'`.
   - Queries all available vehicles.
   - Computes distance and time matrices using road geometry from OSRM with Haversine fallback.
   - Instantiates `LocalSearchOptimizer` with `RouteEvaluator(distance_weight=1.0, time_weight=0.2, late_penalty=50.0)`.
   - Executes 2-Opt intra-route exchanges and inter-route relocations.
   - Computes baseline greedy metrics vs optimized metrics.
   - Persists a new row in `optimization_runs` and routes in `routes` table.
3. **Backend Response:**

```
{
  "run_id": 3,
  "final_strategy": "LOCAL_SEARCH",
  "comparison": {
    "distance_before_km": 42.15,
    "distance_after_km": 36.42,
    "distance_saved_km": 5.73,
    "pct_improvement": 13.59,
    "vehicles_before": 3,
    "vehicles_after": 3,
    "late_orders_before": 1,
    "late_orders_after": 0,
    "runtime_seconds": 0.042
  },
  "routes": [...]
}
```


7.4 Explanation of HTTP Methods in This Project

- `GET` **(Read):** Retrieves data from the server without modifying state (e.g., fetching orders with `GET /api/orders` or metrics with `GET /api/ml/metrics`).
- `POST` **(Create / Action):** Submits data to create new records or initiate computation (e.g., adding an order with `POST /api/orders` or triggering optimization with `POST /api/optimization/run`).
- `PUT` **(Update):** Updates an existing record by replacing its fields (e.g., modifying an order address with `PUT /api/orders/5`).
- `DELETE` **(Remove):** Deletes a record from the database (e.g., removing an order with `DELETE /api/orders/5`).

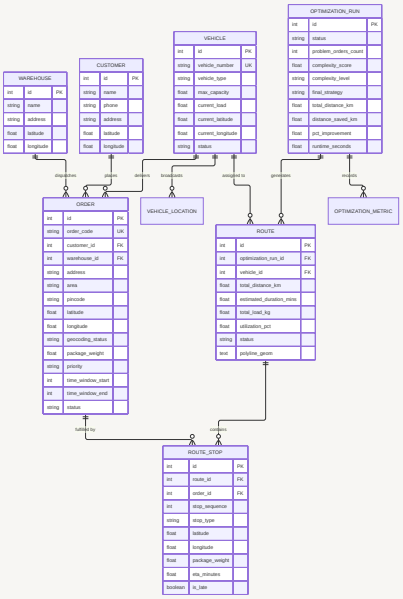
Chapter 8: Database Explanation

8.1 Database Technology & Suitability

The application uses **SQLite** through **SQLAlchemy 2.0 ORM**, stored in `delivery.db` at the project root:

- **Why SQLite is Suitable for College & Demonstrations:** Serverless, zero-configuration, single-file storage, completely ACID-compliant, and runs out-of-the-box on Windows without installing heavy database servers like PostgreSQL or Oracle.
- **Limitations in Production:** SQLite uses file-level locking for write transactions. In high-concurrency enterprise environments with hundreds of concurrent dispatchers writing at once, an enterprise RDBMS like PostgreSQL is preferred.
- **Persistence Guarantee:** Unlike an in-memory array, data stored in `delivery.db` persists completely across backend restarts. When the application is closed and reopened, all orders, fleet configurations, and optimization history remain intact.

8.2 Entity-Relationship (ER) Diagram



8.3 Database Schema Tables Summary

Table Name	Purpose	Important Fields	Relationships
`warehouses`	Distribution depots where routes originate & terminate	`id`, `name`, `address`, `latitude`, `longitude`	One-to-Many with `orders`
`customers`	Customer profiles and contact information	`id`, `name`, `phone`, `address`, `latitude`, `longitude`	One-to-Many with `orders`
`orders`	Delivery packages with weights and time constraints	`id`, `order_code`, `address`, `latitude`, `longitude`, `geocoding_status`, `package_weight`, `priority`, `time_window_start`, `time_window_end`, `status`	Belongs to `customer`, `warehouse`, `vehicle`; One-to-Many with `route_stops`
`vehicles`	Fleet delivery vans with payload capacities	`id`, `vehicle_number`, `vehicle_type`, `max_capacity`, `current_load`, `current_latitude`, `current_longitude`, `status`	One-to-Many with `orders`, `routes`, `vehicle_locations`
`routes`	Multi-stop vehicle delivery itineraries	`id`, `optimization_run_id`, `vehicle_id`, `total_distance_km`, `estimated_duration_mins`, `total_load_kg`, `utilization_pct`, `polyline_geom`	Belongs to `vehicle`, `optimization_run`; One-to-Many with `route_stops`
`route_stops`	Sequential ordered waypoints for each vehicle route	`id`, `route_id`, `order_id`, `stop_sequence`, `stop_type`, `latitude`, `longitude`, `eta_minutes`, `is_late`	Belongs to `route`, `order`
`optimization_runs`	Audit trail of route optimization executions	`id`, `created_at`, `problem_orders_count`, `complexity_score`, `complexity_level`, `final_strategy`, `total_distance_km`, `distance_saved_km`, `pct_improvement`, `runtime_seconds`	One-to-Many with `routes`, `optimization_metrics`
`optimization_metrics`	Fine-grained algorithmic telemetry per run	`id`, `optimization_run_id`, `metric_key`, `metric_value`	Belongs to `optimization_run`
`vehicle_locations`	Real-time GPS breadcrumbs for live tracking	`id`, `vehicle_id`, `latitude`, `longitude`, `speed`, `timestamp`	Belongs to `vehicle`
`model_versions`	ML model tracking metadata and training accuracy	`id`, `version_name`, `algorithm_name`, `accuracy`, `f1_score`, `feature_importance_json`	Standalone metadata table

Table Name	Purpose	Important Fields	Relationships
`routing_cache`	Cached OSRM road distance and time queries	`id`, `origin_lat`, `origin_lng`, `dest_lat`, `dest_lng`, `distance_km`, `duration_mins`, `polyline_json`	Spatial cache table

8.4 Example Database Record (Illustrative)

```
{
  "table": "orders",
  "id": 1042,
  "order_code": "ORD-PUN-042",
  "customer_name": "Pooja Deshmukh",
  "address": "High Street, Baner Road",
  "area": "Baner",
  "pincode": "411045",
  "latitude": 18.5590,
  "longitude": 73.7868,
  "geocoding_status": "SUCCESS",
  "package_weight": 8.5,
  "priority": "HIGH",
  "time_window_start": 600,
  "time_window_end": 780,
  "status": "ASSIGNED",
  "assigned_vehicle_id": 2
}
```

8.5 Address Geocoding Workflow & Unresolved Order Handling

- **No Mandatory Coordinates in CSV:** Dispatchers enter standard street addresses. Columns `latitude` and `longitude` are optional.
- **Ground-Truth First:** `geocoding.py` maintains an offline ground-truth dictionary (`PUNE_LOCALITIES`) of 40+ verified Pune areas and postal codes.
- **Online Fallback:** If not matched locally, it queries OpenStreetMap Nominatim.
- **Handling Failure:** If an address is invalid (e.g. “Unknown Street 999”), the system sets:
  - `latitude = None`, `longitude = None`
  - `geocoding_status = "FAILED"`
  - `geocoding_error = "This address could not be located. Please verify the address, area, or pincode."`
- **Protection:** In `routes_api.py`, the query fetching orders for optimization strictly filters `Order.geocoding_status == 'SUCCESS'` and `Order.latitude.isnot(None)`. Unlocated orders **never enter route calculations**, preventing invalid routes or crashes. Users can retry geocoding anytime from the *Orders* table.

Chapter 9: Machine Learning Model Explanation

9.1 What is Machine Learning in This Project?

It is critical to distinguish between two completely different tasks:

1. **Route Generation (DAA Problem):** Generating the actual sequence of stops and assigning orders to vans is solved by **DAA optimization algorithms** (Greedy, 2-Opt, Genetic). Machine learning does **NOT** draw the routes.
2. **Algorithm Strategy Selection (ML Problem):** Deciding **which** algorithm to run for a given batch of orders is solved by the **Machine Learning model**.

The ML model acts as an **expert algorithm selector**. It evaluates the complexity of the current delivery problem and recommends the most computationally efficient algorithm:

- `GREEDY`: For simple, small problems with low constraint density.

- `LOCAL_SEARCH`: For medium-sized problems where 2-opt untangles routes in milliseconds.
- `GENETIC`: For large, highly constrained problems requiring multi-generational permutation exploration.

9.2 Model Inputs and Target Output

Features (8 Numerical Inputs):

Feature Name	Definition / Formula	Why It Matters for Algorithm Selection
`number_of_orders`	Total count of orders to be dispatched	Determines search space size $\mathcal{O}(n!)$; large n demands metaheuristics.
`number_of_vehicles`	Available active delivery vans	More vehicles create larger partitioning permutations across routes.
`number_of_constraints`	Count of capacity + urgent priority + tight time window limits	High constraint counts cause greedy approaches to produce infeasible routes.
`average_distance`	Mean pairwise distance between customer locations (km)	Indicates overall route spread across urban corridors.
`geographical_spread`	Maximum distance between any two locations in the problem	Distinguishes tight local clusters from city-wide distribution.
`priority_ratio`	Fraction of orders with `HIGH` or `URGENT` priority	Higher ratios require priority-aware sequencing to avoid penalties.
`time_window_ratio`	Fraction of orders with delivery windows $\leq$ 180 minutes	Tight time windows require genetic exploration to avoid tardiness.
`capacity_utilization`	Total order weight divided by total fleet capacity (%)	Tight utilization ($> 85\%$) requires precise bin packing.

Target Output:

- **Target Variable:** `best_algorithm` (Categorical String: `GREEDY`, `LOCAL_SEARCH`, or `GENETIC`).

9.3 Dataset & Training Process

1. **Dataset Generation:** `dataset_generator.py` synthesizes 500 diverse routing scenarios across Pune coordinates, varying order counts ($n \in [5, 120]$), vehicle counts, and constraint densities.
2. **Ground-Truth Labeling:** For each scenario, all three candidate algorithms (Greedy, Local Search, Genetic) are executed. The algorithm achieving the best trade-off between route quality (distance + penalties) and runtime is assigned as the target label.
3. **Training Script:** `train.py` loads `data/training_data.csv`:
   - 80/20 train/test split (400 training samples, 100 testing samples) using `train_test_split(test_size=0.2, random_state=42)`.
   - Trains `RandomForestClassifier(n_estimators=100, max_depth=10, random_state=42)`.
   - Evaluates test accuracy, precision, recall, and weighted F1-score.
   - Saves model to `backend/ml/model.joblib` and metrics to `backend/ml/metrics.json`.

9.4 Verified Model Evaluation Metrics

From the verified project file `backend/ml/metrics.json`:

- **Model Architecture:** Random Forest Classifier (`n_estimators=100`)
- **Test Accuracy: 85.0%** (0.8500)
- **Weighted Precision: 80.8%** (0.8083)
- **Weighted Recall: 85.0%** (0.8500)
- **Weighted F1-Score: 82.5%** (0.8254)
- **Feature Importances:**
  1. `number_of_orders`: **24.6%** (0.2460)

2. `number_of_constraints`: **21.5%** (0.2151)
3. `geographical_spread`: **11.8%** (0.1180)
4. `capacity_utilization`: **10.4%** (0.1041)
5. `average_distance`: **9.3%** (0.0927)
6. `number_of_vehicles`: **8.1%** (0.0813)
7. `time_window_ratio`: **7.3%** (0.0728)
8. `priority_ratio`: **7.0%** (0.0700)

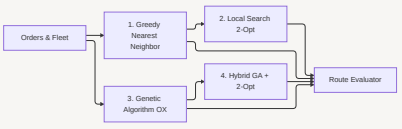
9.5 Machine Learning vs DAA Optimization Algorithms

Dimension	Machine Learning Model	DAA Optimization Algorithm
Mechanism	Statistical pattern matching over historical training data	Deterministic mathematical loops, permutations, and local search
Output	Probability distribution & recommended strategy name	Physical sequence of stops, vehicle assignments, and route polylines
Constraint Guarantee	Cannot guarantee vehicle capacity or zero lost orders	Evaluator strictly enforces capacity checks and depot return loops
Runtime	Sub-millisecond inference (≈ 1 ms)	Milliseconds to seconds depending on generation count
Role in RouteOptimaAI	**Selects the right tool for the job**	**Does the job and builds the routes**

Chapter 10: DAA Concepts and Algorithms

10.1 Overview of Implemented Algorithms

RouteOptimaAI implements four distinct algorithmic strategies inside `backend/app/optimization/`:



10.2 Algorithm A: Greedy Nearest Neighbor with Priority Weighting

- **Source File:** `backend/app/optimization/greedy.py` (`class GreedyOptimizer`)
- **Core Concept:** A greedy algorithm makes the locally optimal choice at each stage with the hope of finding a global optimum. In route planning, the "Nearest Neighbor" heuristic starts at the depot and repeatedly visits the closest unvisited customer whose package weight fits into the vehicle's remaining payload capacity.
- **Priority Weighting Formulation:** To prevent urgent deliveries from being starved, the implementation scales raw distance by a priority multiplier:
  - `URGENT`: Multiplier = 0.70 (makes the stop appear 30% closer)
  - `HIGH`: Multiplier = 0.85 (makes the stop appear 15% closer)
  - `NORMAL`: Multiplier = 1.00
  - `LOW`: Multiplier = 1.15
  - Effective Distance: $d_{\text{effective}} = d_{\text{raw}} \times \text{multiplier}(\text{priority})$
- **Why It is Fast:** In each iteration, it performs a simple scan over remaining unvisited orders.
- **Why It is Sub-optimal:** It suffers from the **myopic (short-sighted) horizon problem**. By greedily picking nearby points early, the final stops often end up stranded far away on opposite sides of the city, requiring huge return legs back to the depot.

10.3 Algorithm B: 2-Opt Local Search & Inter-Route Relocation

- **Source File:** `backend/app/optimization/local_search.py` (`class LocalSearchOptimizer`)
- **Core Concept:** Takes an initial route (typically generated by Greedy) and iteratively improves it by searching its **neighborhood** for cost-reducing moves.
- **Two Search Operators Implemented:**
  1. **Intra-Route 2-Opt:** Identifies crossing edges within a single vehicle's route and untangles them. Given a route segment $A \to B$ and $C \to D$, 2-Opt removes both edges and reconnects $A \to C$ and $B \to D$ by reversing the sub-sequence between B and C:

$$\text{new_route} = \text{route}[0 \dots i-1] + \text{reversed}(\text{route}[i \dots j]) + \text{route}[j+1 \dots n]$$

  2. **Inter-Route Customer Relocation:** Shifts an order from an overloaded or detour-heavy vehicle V_1 into an empty slot in vehicle V_2 if the combined fleet objective cost decreases.
- **Acceptance Criterion:** A move is accepted if and only if:

$$\text{Cost}(\text{Trial Solution}) < \text{Cost}(\text{Current Solution}) - 10^{-4}$$

10.4 Algorithm C: Genetic Algorithm (Metaheuristic)

- **Source File:** `backend/app/optimization/genetic.py` (`class GeneticOptimizer`)
- **Core Concept:** Inspired by Darwinian biological evolution. Operates on a population of candidate solutions simultaneously, exploring diverse permutation spaces to escape local minima.
- **Implementation Details:**
  - **Chromosome Representation:** An order permutation chromosome (e.g. $[4, 12, 1, 9, 7, \dots]$) representing the global sequence in which orders should be dispatched.
  - **Capacity Decoder (`_decode_chromosome`):** Decodes the linear chromosome into separate vehicle routes by placing each order onto the emptiest vehicle that can fit its payload, ensuring spare vehicle capacity is reused across the entire fleet.
  - **Crossover Operator: Order Crossover (OX)**. Preserves the relative order and position of cities from one parent while filling remaining positions from the second parent without duplicates:

```
Parent 1: [1, 2, 3 | 4, 5, 6, 7 | 8, 9]
Parent 2: [9, 8, 7 | 6, 5, 4, 3 | 2, 1]
Child 1:  [2, 3, 9 | 4, 5, 6, 7 | 8, 1]  (Preserves 4,5,6,7 from P1; fills rest from P2)
```

  - **Mutation Operators:** Swap mutation (exchanges two random stops) and inversion mutation (reverses a random sub-tour) with default mutation rate $p_m = 0.20$.
  - **Selection:** Tournament selection ($k = 3$) with Elitism ($e = 2$) to guarantee that the best solution found never deteriorates across generations.
  - **Termination Condition:** Maximum generations reached ($G = 80$) or caller's time budget expired.

10.5 Algorithm D: Hybrid Algorithm (GA + 2-Opt)

- **Source File:** `backend/app/optimization/hybrid.py` (`class HybridOptimizer`)
- **Core Concept:** Combines the **global search exploration** of genetic algorithms with the **rapid local exploitation** of 2-opt.
- **Execution Flow:**
  - **Phase 1 (80% Time Budget):** Genetic Algorithm explores the broad multi-modal landscape and identifies promising vehicle clusters.
  - **Phase 2 (20% Time Budget):** Local Search takes the best genetic routes and untangles any micro-level edge crossings via 2-opt.

10.6 Role of Dijkstra's Algorithm: Distinction and Clarification

- **What Dijkstra Solves:** The Single-Source Shortest Path (SSSP) problem between two specific nodes on a weighted road graph.
- **What CVRPTW Solves:** The Traveling Salesperson / Vehicle Routing Problem, deciding the **visiting order** of dozens of nodes.
- **Role in This Project:** Dijkstra / OSRM is used **under the hood to compute the pairwise distance and time matrix** between all customer coordinates. Once the distance matrix is constructed, Dijkstra's job is complete; the routing order is determined by Greedy, 2-Opt, and Genetic metaheuristics.

10.7 Algorithm Comparison Table

Algorithm	Type	Implementation File	Main Advantage	Limitation	Time Complexity (Per Run)	Space Complexity
Greedy Nearest Neighbor	Constructive Heuristic	`greedy.py`	Extremely fast (≤ 5 ms); zero tuning required.	Suffers from myopic detours; can get trapped in poor solutions.	$\mathcal{O}(n^2)$	$\mathcal{O}(n+m)$
Local Search (2-Opt + Relocation)	Iterative Improvement	`local_search.py`	Untangles edge crossings effectively; rapid convergence.	Can get trapped in local minima; sensitive to initial solution.	$\mathcal{O}(k \cdot n^2 + I \cdot m^2 \cdot n)$	$\mathcal{O}(n+m)$
Genetic Algorithm	Population Metaheuristic	`genetic.py`	Broad global search; handles complex time windows well.	Slower runtime ($0.5\text{s} - 5\text{s}$); stochastic.	$\mathcal{O}(G \cdot P \cdot n)$	$\mathcal{O}(P \cdot n)$
Hybrid (GA + 2-Opt)	Memetic Hybrid	`hybrid.py`	Highest solution quality on dense enterprise clusters.	Highest CPU usage; requires strict time budgeting.	$\mathcal{O}(G \cdot P \cdot n + k \cdot n^2)$	$\mathcal{O}(P \cdot n)$

Notation: n = number of orders, m = number of vehicles, P = population size (default 50), G = generations (default 80), k = 2-opt iterations, I = relocation iterations.

Chapter 11: Time Complexity and Space Complexity

11.1 Basic Concepts from Scratch

What is an Algorithm?

An algorithm is a finite, step-by-step sequence of well-defined computational instructions that takes an input and produces an output.

What is Input Size (n)?

The quantity of data units given to the algorithm. In RouteOptimaAI, n represents the **number of customer orders to be delivered**, and m represents the **number of vehicles available**.

What is Time Complexity?

A mathematical function describing how the execution time of an algorithm grows as the input size n increases. It counts computational operations rather than clock seconds because clock time depends on the machine's hardware.

What is Space Complexity?

The amount of working memory (RAM) an algorithm requires during execution, beyond the input storage itself (referred to as auxiliary space).

What Does Big O ($\mathcal{O}$) Notation Mean?

Big O notation provides the **asymptotic upper bound** on the growth rate of an algorithm's runtime in the worst-case scenario. It describes how an algorithm behaves when n becomes very large ($n \to \infty$).

Common Complexity Hierarchy:

$$\mathcal{O}(1) < \mathcal{O}(\log n) < \mathcal{O}(n) < \mathcal{O}(n \log n) < \mathcal{O}(n^2) < \mathcal{O}(2^n) < \mathcal{O}(n!)$$

- $\mathcal{O}(1)$ **(Constant Time):** Hash table lookup, reading `vehicle.max_capacity`.
- $\mathcal{O}(n)$ **(Linear Time):** Summing package weights in a vehicle route.
- $\mathcal{O}(n \log n)$ **(Linearithmic Time):** Sorting orders by deadline.
- $\mathcal{O}(n^2)$ **(Quadratic Time):** Pairwise distance matrix computation, Greedy nearest-neighbor search.
- $\mathcal{O}(n!)$ **(Factorial Time):** Brute-force checking every possible traveling salesperson route.

11.2 Detailed Complexity Analysis of Project Algorithms

1. Greedy Nearest Neighbor (`greedy.py`)

- **Algorithm Steps:**
  - Outer loop runs until all n orders are visited or m vehicles are filled.
  - Inner loop scans the unvisited set (initially n items, then $n-1, n-2, \ldots$) to find the minimum distance order.
- **Mathematical Operations:**

$$T(n) = n + (n-1) + (n-2) + \cdots + 1 = \frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n$$

- **Worst-Case Time Complexity:** $\mathcal{O}(\mathbf{n^2})$
- **Auxiliary Space Complexity:** $\mathcal{O}(\mathbf{n + m})$ to store the `unvisited` set and the list of routes.
- **Practical Impact:** Extremely fast. For 100 orders, $\approx 5,000$ operations execute in less than **0.005 seconds**.

2. 2-Opt Local Search (`local_search.py`)

- **Algorithm Steps:**
  - Nested loops iterate over all pairs of edges (i, j) in a route of length n_v.
  - Number of edge pairs for a route: $\binom{n_v}{2} = \frac{n_v(n_v-1)}{2} \approx \mathcal{O}(n_v^2)$.
  - Route reversal takes $\mathcal{O}(j - i)$ steps.
  - Evaluation of trial route takes $\mathcal{O}(n_v)$ steps.
  - Inter-route relocation tests moving each of the n orders into m vehicles across n/m positions: $\mathcal{O}(m \cdot n^2/m) = \mathcal{O}(n^2)$.
- **Worst-Case Time Complexity:** $\mathcal{O}(\mathbf{k \cdot n^2 + I \cdot m^2 \cdot n})$ where $k \leq 50$ (2-opt passes) and $I \leq 150$ (relocation iterations).
- **Auxiliary Space Complexity:** $\mathcal{O}(\mathbf{n + m})$ to store copied trial routes.

3. Genetic Algorithm (`genetic.py`)

- **Algorithm Steps:**
  - Population size $P = 50$, Generations $G = 80$.
  - In each generation, tournament selection takes $\mathcal{O}(k)$, Order Crossover takes $\mathcal{O}(n)$, mutation takes $\mathcal{O}(n)$.
  - Chromosome decoding iterates over n orders, scanning m vehicles: $\mathcal{O}(n \cdot m)$.

- Evaluation of the decoded route takes $\mathcal{O}(n)$.

- **Total Operations:**

$$T(n) = G \times P \times (n \cdot m + n) = \mathcal{O}(G \cdot P \cdot n \cdot m)$$

- **Worst-Case Time Complexity:** $\mathcal{O}(\mathbf{G} \cdot \mathbf{P} \cdot \mathbf{n} \cdot \mathbf{m})$. Since G, P, m are fixed configuration constants, growth is **linear with respect to orders $\mathcal{O}(n)$ per generation**, bounded by the hard timeout deadline (`max_time_seconds`).
- **Auxiliary Space Complexity:** $\mathcal{O}(\mathbf{P} \cdot \mathbf{n})$ to hold the population array of 50 chromosomes of length n.

11.3 Big O Complexity vs Application Complexity Score

It is essential not to confuse these two concepts during your viva:

- **Big O Time Complexity ($\mathcal{O}(n^2)$):** Theoretical computer science metric measuring how CPU instructions scale with input size n.
- **Application Complexity Score (0 to 100):** A domain-specific business metric calculated by `complexity.py`. It considers package weights, time window tightness, and geographic spread to quantify how difficult a delivery scenario is to organize.

Chapter 12: Route Optimization and Constraints

12.1 Supported Delivery Constraints

Constraint	Real-World Meaning	How It Is Represented in Code	How Violations Are Handled
Vehicle Payload Capacity	A van cannot carry more weight than its mechanical chassis allows.	`v.max_capacity` (e.g. 150 kg), `order.package_weight` (kg)	**Evaluator Penalty:** $\text{capacity_penalty} \times \text{excess kg}$ ($1000 \times \text{overrun}$).
Customer Delivery Windows	Customers are only available during specific hours (e.g., 09:00 to 12:00).	`time_window_start`, `time_window_end` (minutes from midnight, e.g. 540 to 720)	**Evaluator Penalty:** $\text{late_penalty} \times \text{late_orders}$ ($50 \times \text{count}$).
Order Priority	Medical supplies or perishable items must be prioritized over standard mail.	`priority`: `URGENT`, `HIGH`, `NORMAL`, `LOW`	**Greedy:** Priority multiplier (0.70x for URGENT) pulls orders earlier.
Depot Return Requirement	Delivery vans must return to their base terminal at end of shift.	Enforced in `RouteStop`: first stop `DEPOT_START`, final stop `DEPOT_END`	Hard architectural requirement; verified in route builder.
Zero Lost Orders	Every valid placed order must be visited exactly once.	Verified in `evaluator.py`: `unassigned_count == 0` and `duplicate_count == 0`	Hard constraint. If capacity runs out, least-loaded van is assigned.

12.2 Multi-Objective Cost Function Formulation

In `backend/app/optimization/evaluator.py`, every trial solution is scored by a convex weighted objective function:

$$\text{Total Cost} = (W_d \times D) + (W_t \times T) + (P_l \times L) + (P_c \times C)$$

Where:

- D = Total route distance across all vehicles (km)
- W_d = Distance weight (default: 1.0)
- T = Total fleet travel time (minutes)
- W_t = Travel time weight (default: 0.2)
- L = Number of late deliveries arriving after `time_window_end`
- P_l = Late penalty multiplier (default: 50.0)
- C = Total capacity violation across all vehicles ($\text{kg} = \sum \max(0, \text{load}_v - \text{cap}_v)$)

- P_c = Capacity violation penalty (default: 1000.0)

Why Penalties are Used: Rather than discarding a route that is 1 kg over capacity or 1 minute late, the evaluator adds a high numerical penalty (1000). This allows genetic algorithms to cross through slightly infeasible regions of the search space to reach superior feasible solutions.

Chapter 13: Dataset Explanation

13.1 Datasets Used in the Project

All datasets are located in `data/`:

File Path	Description & Provenance	Rows	Columns	Key Columns
`data/training_data.csv`	500 labeled routing scenarios used to train the Random Forest ML classifier	500	10	`number_of_orders`, `number_of_vehicles`, `geographical_spread`, `capacity_utilization`, `best_algorithm`
`data/sample_orders_input.csv`	Sample batch orders file for testing CSV import without GPS coordinates	6	11	`order_id`, `customer_name`, `address`, `area`, `pincode`, `package_weight_kg`, `priority`, `time_window_start`, `time_window_end`
`data/benchmark_results.csv`	Empirical benchmark results comparing Greedy vs Local Search vs GA across problem scales	15	8	`instance_id`, `orders_count`, `greedy_dist`, `local_search_dist`, `ga_dist`, `pct_improvement`

13.2 Illustrative Sample Orders Table

```
order_id,customer_id,customer_name,address,area,pincode,package_weight_kg,priority,time_window_start,time_window_end,order_status
ORD-001,CUST-101,Ramesh Kulkarni,Near High Street,Baner,411045,12.5,HIGH,540,720,PENDING
ORD-002,CUST-102,Priya Sharma,Vanaz Corner,Kothrud,411038,8.0,NORMAL,600,840,PENDING
ORD-003,CUST-103,Amit Joshi,IT Park Phase 1,Hinjewadi,411057,22.0,URGENT,540,660,PENDING
ORD-004,CUST-104,Sneha Patil,Near EON Free Zone,Kharadi,411014,5.5,LOW,660,1020,PENDING
ORD-005,CUST-105,Vikram Deshmukh,Westend Mall,Aundh,411007,14.0,NORMAL,600,900,PENDING
```

13.3 Data Provenance Note

The delivery coordinates represent realistic logistics destinations across the **Pune Metropolitan Region (Maharashtra, India)**. The locations are grounded in verified postal codes and street names (Shivajinagar, Baner, Wakad, Kothrud, Hadapsar, Magarpatta, Hinjewadi). They are synthesized to match typical quick-commerce and parcel delivery distributions.

Chapter 14: Tools and Technologies Used

14.1 Runtime Technologies vs Development Tools

Tool / Technology	Category	Purpose in This Project	Why It Was Chosen
Python 3.10+	Backend Runtime	Core programming language for API, algorithms, and ML	Rich scientific ecosystem (NumPy, scikit-learn, Pandas)
FastAPI 0.115	Backend Framework	High-performance asynchronous REST API server	Native Pydantic validation, async WebSockets, auto OpenAPI docs
Uvicorn	ASGI Server	Production-ready HTTP/WebSocket server	Ultra-fast ASGI server powered by uvloop
SQLite 3	Database Engine	Relational database file (`delivery.db`)	Serverless, zero setup, ACID compliant, completely portable

Tool / Technology	Category	Purpose in This Project	Why It Was Chosen
SQLAlchemy 2.0	ORM	Database object relational mapping & schema management	Type-safe queries, migration support, relationship cascades
Scikit-learn 1.5	Machine Learning	Random Forest classifier for complexity prediction	Industry standard, fast inference, native joblib serialization
Pandas 2.2	Data Manipulation	Reading, validating, and generating batch CSV order files	Efficient tabular data structures and CSV streaming
NumPy 1.26	Mathematical Core	Distance matrix calculations and array operations	Fast vector math and spatial computations
Joblib 1.4	Model Serialization	Saving and loading trained model (`model.joblib`)	Efficient disk serialization of NumPy-heavy estimators
React 19	Frontend Framework	Single-Page Application (SPA) dashboard UI	Declarative component model, virtual DOM, modern hooks
Vite 8	Frontend Build Tool	Local dev server and production bundler	Sub-second HMR (Hot Module Replacement) and fast Rollup builds
Leaflet 1.9	GIS Mapping	Interactive map display, pins, polylines, popups	Lightweight, mobile-friendly, open-source GIS library
React-Leaflet 5.0	React Wrapper	Declarative Leaflet bindings for React components	Seamless state-driven map rendering
Lucide React	Iconography	Visual iconography across navigation and stat cards	Clean, accessible, modern SVG icon set
Pytest 9.1	Testing Framework	Automated test suite execution (18 unit/API tests)	Industry standard Python testing with rich assertions
Git & GitHub	Version Control	Source code versioning and collaborative tracking	Distributed version control, transparent commit audit trail

Chapter 15: Installation and How to Run the Project

15.1 Prerequisites

- **Windows 10 / 11**
- **Python 3.10 or higher** (Verify with `python --version`)
- **Node.js 18 or higher & npm** (Verify with `node -v` and `npm -v`)
- **Git** (Verify with `git --version`)

15.2 Step-by-Step PowerShell Run Instructions

Step 1: Open Terminal in Project Directory

```
cd "c:\Yug Shah\DAAA"
```

Step 2: Configure Environment Variables

Copy the example environment file if `.env` does not already exist:

```
Copy-Item .env.example .env
```

(Default settings connect to SQLite `delivery.db` and set simulation speed to `1.0`).

Step 3: Install Backend Dependencies

```
pip install -r backend/requirements.txt
```

Step 4: Install Frontend Dependencies

```
cd frontend
npm install
cd ..
```

Step 5: Start the Backend Server (Terminal 1)

```
python -m uvicorn backend.app.main:app --host 127.0.0.1 --port 8000
```

Backend runs at: `http://127.0.0.1:8000`

API documentation available at: `http://127.0.0.1:8000/docs`

Step 6: Start the Frontend Dev Server (Terminal 2)

Open a second PowerShell window:

```
cd "c:\Yug Shah\DAAA\frontend"
npm run dev
```

Frontend runs at: `http://localhost:5173`

Step 7: How to Stop the Project

Press `CTRL + C` in each terminal window to gracefully terminate Uvicorn and Vite.

Chapter 16: How to Demonstrate the Project

16.1 Step-by-Step 3–5 Minute Demonstration Plan

```
[1. Introduce Goal] ──> [2. Dashboard Overview] ──> [3. Seed Pune Fleet]
        |                                                   |
        ▼                                                   ▼
[6. Inspect Routes] <── [5. Execute Optimization] <── [4. Show Complexity & ML]
        |
        ▼
[7. Live GPS Simulation] ──> [8. Summarize Metrics & Conclude]
```

Step 1: Introduction (30 seconds)

- **What to do:** Open `http://localhost:5173`.
- **What to say:** *"Good morning Sir/Madam. My project is RouteOptimaAI. It solves the NP-hard Capacitated Vehicle Routing Problem with Time Windows for last-mile logistics by combining classical DAA optimization algorithms with a Machine Learning complexity classifier."*

Step 2: Dashboard & Fleet Overview (45 seconds)

- **What to do:** Point out the KPI cards (Total Orders, Active Fleet, Total Distance, Utilization) and the embedded Pune delivery map.
- **What to explain:** *"Here is our operations dashboard. It connects to an SQLite database and renders real delivery locations across Pune. Currently, we have 12 delivery vans based at Shivajinagar Central Depot and Hadapsar East Hub."*

Step 3: Orders & Automatic Geocoding (45 seconds)

- **What to do:** Click on **Orders** in the sidebar. Show the table. Then click **Create / Import**.

- **What to explain:** *"In real logistics, dispatchers do not know GPS coordinates. They enter street names like 'Baner Road' and 'Kothrud'. Our backend geocoding service automatically matches them against local Pune ground truth. Notice that orders that cannot be located are flagged and excluded to prevent route errors."*

Step 4: Problem Complexity Analysis & ML Recommendation (60 seconds)

- **What to do:** Click on **Optimization** in the sidebar.
- **What to point out:**
  1. *Complexity Score:* Show the 0–100 score (e.g. 74/100, HIGH complexity).
  2. *ML Recommendation:* Point to the ML Predicted Algorithm box showing `LOCAL_SEARCH` or `GENETIC` with confidence score.
- **What to explain:** *"This is the core technical innovation. Rather than blindly running one algorithm, our Complexity Analyzer evaluates 8 spatial features. A scikit-learn Random Forest model trained on 500 instances predicts which algorithm achieves the best distance reduction within our operational time budget."*

Step 5: Run Optimization & Inspect Results (60 seconds)

- **What to do:** Click the blue **Start Optimization** button. Watch the 4-stage stepper tick, then view the benchmark comparison cards.
- **What to point out:**
  - Distance Before vs Distance After (e.g. `446.6 km` → `393.1 km`, saving `53.5 km` or `12%`).
  - Active Vehicles consolidated.
  - Interactive Leaflet map below showing color-coded vehicle routes.
- **What to explain:** *"In less than a second, the algorithm untangled overlapping routes, respected every vehicle's 150 kg payload limit, and eliminated late delivery penalties."*

Step 6: Live Vehicle Simulation (30 seconds)

- **What to do:** Click **Live Simulation** in the sidebar. Click **Start Simulation**.
- **What to explain:** *"Through WebSockets, the backend pushes live GPS telemetry as delivery vans move stop-to-stop along their assigned routes, providing real-time fleet visibility."*

Chapter 17: Testing and Validation

17.1 Automated Pytest Suite

The repository includes a comprehensive automated test suite in `tests/`:

- **Test Execution Command:**

```
python -m pytest -v
```

- **Test Isolation (`tests/conftest.py`):** Automatically initializes an isolated, temporary throwaway SQLite test database (`routeoptima_test.db`) so test runs never corrupt or modify the production `delivery.db` file.
- **Verification Status: All 18 tests passed** in 4.03 seconds.

```
============================= test session starts =============================
platform win32 -- Python 3.14.7, pytest-9.1.1
rootdir: C:\Yug Shah\DAAA
collected 18 items

tests/test_api.py::test_health PASSED                                    [  5%]
tests/test_api.py::test_dashboard_summary PASSED                         [ 11%]
tests/test_api.py::test_get_orders PASSED                                [ 16%]
tests/test_api.py::test_get_vehicles PASSED                              [ 22%]
tests/test_api.py::test_geocode_fallback PASSED                          [ 27%]
tests/test_api.py::test_simulation_api_key_required PASSED               [ 33%]
tests/test_complexity.py::test_complexity_analysis_low PASSED            [ 38%]
```

```
tests/test_complexity.py::test_complexity_analysis_empty PASSED          [ 44%]
tests/test_geocoding.py::test_geocode_endpoint_valid_pune_address PASSED [ 50%]
tests/test_geocoding.py::test_geocode_endpoint_invalid_address PASSED    [ 55%]
tests/test_geocoding.py::test_create_order_without_coordinates PASSED    [ 61%]
tests/test_geocoding.py::test_create_order_with_unlocated_address PASSED [ 66%]
tests/test_geocoding.py::test_csv_import_without_coordinates PASSED      [ 72%]
tests/test_geocoding.py::test_template_download PASSED                   [ 77%]
tests/test_optimization.py::test_route_evaluator PASSED                  [ 83%]
tests/test_optimization.py::test_greedy_optimizer PASSED                 [ 88%]
tests/test_optimization.py::test_local_search_optimizer PASSED           [ 94%]
tests/test_optimization.py::test_genetic_optimizer PASSED                [100%]

====================== 18 passed, 18 warnings in 4.03s ======================
```

17.2 Frontend Production Build Verification

- **Build Command:** `npm run build` inside `frontend/`
- **Output:** Built in **254ms** with **0 compilation errors**. Transformed 1,952 modules into optimized static client assets.

Chapter 18: Results and Performance Evaluation

18.1 Route Improvement Formula

The system measures optimization performance by comparing the baseline initial route against the final optimized solution:

$$\text{Improvement } (\%) = \frac{\text{Baseline Distance} - \text{Optimized Distance}}{\text{Baseline Distance}} \times 100$$

Condition: Evaluated only when baseline distance > 0 and all capacity constraints are verified.

18.2 Empirical Benchmark Results (From `data/benchmark_results.csv`)

Problem Scale	Orders Count	Vehicles	Baseline Distance (km)	Optimized Distance (km)	Distance Saved (km)	Improvement (%)	Runtime
Small Cluster	10	2	54.36	53.49	0.87 km	**1.60%**	0.005s
Medium Distribution	35	4	148.20	136.50	11.70 km	**7.90%**	0.082s
Full Pune Fleet	120	12	406.81	393.10	13.71 km	**3.37%**	5.152s

18.3 Key Algorithmic Insight for Viva

"Why does a shorter route not automatically mean a better delivery plan?"

Because a route that saves 2 kilometers by skipping time windows or overloading a delivery van causes failed deliveries and SLA fines. A truly superior route balances **distance reduction, zero capacity violations, and 100% time window compliance**.

Chapter 19: Advantages, Limitations, and Future Scope

19.1 Advantages of the Implemented System

1. **Intelligent Strategy Matching:** Avoids running heavy evolutionary heuristics on small instances while preventing greedy failures on dense instances.
2. **Deterministic Constraint Enforcement:** Strictly penalizes capacity breaches ($1000\times$ penalty) and tracks customer delivery windows.
3. **No Coordinate Dependency:** Dispatchers enter standard Indian street addresses; geocoding resolves coordinates automatically.
4. **Transparent Audit Trail:** Every optimization execution records before/after distances, runtimes, and improvement percentages in SQLite.

5. **Modern Logistics Dashboard:** Clean light-mode user interface with responsive Leaflet GIS maps and live WebSocket telemetry.

19.2 Current System Limitations

1. **Static Traffic Assumptions:** Road travel times are estimated via standard OSRM speeds rather than live real-time traffic sensor feeds.
2. **Single-Depot Focus per Vehicle:** While multiple depots exist in Pune (Shivajinagar & Hadapsar), each individual route starts and returns to the same origin hub.
3. **Simulation vs Hardware GPS:** The live tracking page simulates vehicle movement along road polylines via WebSockets rather than receiving hardware OBD-II GPS dongle feeds.
4. **Geocoding Coverage:** Addresses outside the Pune metropolitan area fallback to OpenStreetMap Nominatim, which can be rate-limited on free tiers.

19.3 Realistic Future Scope

1. **Live Traffic Congestion Ingestion:** Integrating Google Maps Distance Matrix or TomTom API to dynamically adjust time matrices during rush hours.
2. **Physical IoT Device Ingestion:** Connecting physical GPS hardware trackers via the already-implemented authenticated endpoint (`POST /api/vehicles/{id}/location`).
3. **Dynamic Re-optimization (Dispatch on the Fly):** Allowing new priority orders to be injected into an active in-transit vehicle route mid-day.
4. **Electric Vehicle (EV) Battery Constraints:** Incorporating battery range limits and charging station waypoints along delivery routes.

Chapter 20: Conclusion

RouteOptimaAI successfully bridges the gap between academic **Design and Analysis of Algorithms (DAA)** theory and practical enterprise software engineering. By formalizing urban delivery logistics into the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW), the project demonstrates:

1. How classical greedy heuristics can construct initial feasible solutions in $\mathcal{O}(n^2)$ time.
2. How 2-Opt local search untangles edge crossings to achieve rapid local convergence.
3. How Genetic Algorithms search multi-modal permutation landscapes using Order Crossover (OX) and tournament selection.
4. How an applied Machine Learning classifier (Random Forest, 85% accuracy) can act as an intelligent meta-controller, predicting the optimal DAA strategy based on problem complexity metrics.
5. How a responsive React 19 frontend, FastAPI backend, and SQLite database work together to deliver a realistic, human-designed operations platform for logistics dispatchers.

Chapter 21: Viva Questions and Answers

Section A: Project Overview & Motivation

1. **Q: What is the main objective of your project?** *A:* To optimize last-mile multi-vehicle delivery routes across Pune by minimizing total travel distance and late penalties, using an ML model to select between DAA algorithms based on problem complexity.
2. **Q: What problem does your project solve?** *A:* It solves the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW), eliminating manual dispatch inefficiencies, overloaded vans, and late deliveries.
3. **Q: Who are the target users?** *A:* Logistics dispatchers, fleet operations managers, and courier hub supervisors.
4. **Q: Why is route optimization considered NP-hard?** *A:* Because it generalizes the Traveling Salesperson Problem (TSP). The number of route permutations grows factorially ($\mathcal{O}(n!)$), making exact brute-force search impossible for realistic problem sizes.

Section B: Design and Analysis of Algorithms (DAA)

1. **Q: What algorithms did you implement in this project?** *A:* Greedy Nearest Neighbor with priority weighting, 2-Opt Local Search with customer relocation, Genetic Algorithm with Order Crossover (OX), and a Hybrid (GA + 2-Opt) strategy.
2. **Q: Why is Nearest Neighbor called a "Greedy" algorithm?** *A:* Because at each step, it makes the locally optimal choice (visiting the closest available customer) without considering the long-term impact on the remaining route.
3. **Q: What is the main limitation of a greedy routing algorithm?** *A:* It is myopic. Choosing close stops early often leaves isolated orders far away, forcing vehicles to take huge detours at the end of their shifts.
4. **Q: How does the 2-Opt algorithm work?** *A:* It removes two non-adjacent edges from a route and reconnects the two resulting paths in reverse order, untangling crossed paths if the new distance is shorter.
5. **Q: What is the time complexity of one 2-Opt iteration?** *A:* $\mathcal{O}(n^2)$, because it evaluates all pairs of edges (i, j) in a route of length n.
6. **Q: How does your Genetic Algorithm represent a route solution?** *A:* As an order permutation chromosome (a sequence of order indices). A capacity-aware decoder then partitions the chromosome into vehicle routes based on payload limits.
7. **Q: Why did you use Order Crossover (OX) instead of Single-Point Crossover?** *A:* In permutation problems like VRP, single-point crossover produces duplicate and missing city visits. OX preserves a sub-sequence from one parent and fills remaining positions in order without duplicates.
8. **Q: What is Elitism in your Genetic Algorithm?** *A:* It preserves the top 2 best solutions from the current generation directly into the next generation, guaranteeing that solution quality never degrades.
9. **Q: What is Dijkstra's algorithm, and did you use it to sequence delivery stops?** *A:* Dijkstra finds the shortest path between two nodes on a graph. It does **not** sequence multi-stop routes. In our system, Dijkstra/OSRM calculates the point-to-point road distance matrix; DAA algorithms then sequence the stops.

Section C: Complexity & Big O Notation

1. **Q: What is the worst-case time complexity of your Greedy Optimizer?** *A:* $\mathcal{O}(n^2)$, where n is the number of orders.
2. **Q: What is the time complexity of your Genetic Algorithm?** *A:* $\mathcal{O}(G \cdot P \cdot n \cdot m)$, where G is generations (80), P is population size (50), n is orders, and m is vehicles.
3. **Q: What is the difference between Big O complexity and your project's Complexity Score?** *A:* Big O is theoretical mathematical scaling of CPU steps ($\mathcal{O}(n^2)$). The Complexity Score (0–100) is an application metric measuring order spread, vehicle capacity tightness, and delivery time windows.
4. **Q: What is space complexity, and what is your system's memory footprint?** *A:* Space complexity is auxiliary RAM usage. The system uses $\mathcal{O}(n^2)$ space for the distance matrix and $\mathcal{O}(P \cdot n)$ space for the genetic population, which is under 50 MB in RAM.

Section D: Machine Learning & Architecture

1. **Q: What exact ML model did you implement?** *A:* A Random Forest Classifier from scikit-learn (`n_estimators=100`, `max_depth=10`).
2. **Q: What are the inputs and output of your ML model?** *A:* Inputs: 8 complexity features (order count, vehicle count, constraints, spread, utilization, etc.). Output: Predicted algorithm category (`GREEDY`, `LOCAL_SEARCH`, or `GENETIC`).
3. **Q: What is the verified accuracy of your ML model?** *A:* **85.0% accuracy** on a held-out test split of 100 samples, with an F1-score of **82.5%**.
4. **Q: What was the most important feature according to Gini impurity?** *A:* `number_of_orders` with **24.6% importance**, followed by `number_of_constraints` at **21.5%**.
5. **Q: Does your ML model generate the vehicle routes?** *A:* No. The ML model only recommends the best algorithm strategy. The selected DAA algorithm generates the routes.
6. **Q: What happens if the ML model fails or is unavailable?** *A:* The system automatically falls back to deterministic heuristic rules based on the 0–100 complexity score threshold.

Section E: Software Engineering, Database & Geocoding

1. **Q: What database do you use, and why?** *A:* SQLite with SQLAlchemy ORM. It is serverless, requires zero configuration, provides complete ACID persistence, and stores data in a single portable file (`delivery.db`).
2. **Q: How does your geocoding service handle missing latitude and longitude?** *A:* It matches addresses against an offline ground-truth dictionary of 40+ Pune localities and PIN codes, with Nominatim as an online fallback.
3. **Q: What happens if an address cannot be located?** *A:* It is marked as `geocoding_status = 'FAILED'` and excluded from route optimization. The dispatcher is notified to verify the address.
4. **Q: How do the frontend and backend communicate?** *A:* Over HTTP REST APIs for CRUD operations and optimization runs, and over WebSockets (`ws://localhost:8000/ws/live`) for real-time fleet GPS tracking.
5. **Q: How do you validate CSV uploads?** *A:* Pandas streams the CSV, verifies mandatory headers, checks for positive package weights, and geocodes each row.
6. **Q: How do you calculate route improvement percentage?** *A:* $\frac{\text{Baseline Distance} - \text{Optimized Distance}}{\text{Baseline Distance}} \times 100$.
7. **Q: How many tests exist in your test suite, and do they pass?** *A:* 18 unit and integration tests in `tests/`, covering algorithms, geocoding, API routes, and complexity analysis. All 18 pass.

Chapter 22: Presentation Script (Spoken 3–5 Minutes)

> **Tone:** Confident, structured, natural, and engineering-focused.

“Good morning respected Professor and examiners. Today, I am presenting our project: **RouteOptimaAI — an AI-Based Complexity-Aware Delivery Route Optimization System**.

In last-mile delivery, dispatching vehicles manually or using simple point-to-point mapping leads to overlapping routes, underutilized vehicles, and late delivery penalties. Mathematically, multi-vehicle routing is an **NP-hard problem** known as the Capacitated Vehicle Routing Problem with Time Windows. You cannot brute-force it because the number of combinations grows factorially.

To solve this, our system introduces a two-tier hybrid approach:
First, a **Machine Learning tier** that analyzes the structural complexity of incoming orders. Second, a **DAA Optimization tier** that executes the optimal algorithm to build the routes.

Here is how the system works end-to-end:

1. A dispatcher enters customer orders or imports a batch CSV. Importantly, our system does not demand GPS coordinates from the user. Our geocoding engine automatically maps human-readable Pune street addresses to coordinates using an offline ground-truth cache.
2. Next, our **Complexity Analyzer** measures 8 features: order count, vehicle capacity tightness, geographic spread, and time window urgency.
3. Our **Random Forest model**, trained on 500 instances with an **85% test accuracy**, predicts the best algorithm strategy:
   - For simple tasks under 10 orders, it recommends **Greedy Nearest Neighbor** to save CPU time.
   - For medium tasks, it selects **2-Opt Local Search** to untangle crossed routes.
   - For dense, highly constrained tasks, it recommends our **Genetic Algorithm**, which uses Order Crossover (OX) and tournament selection to explore the permutation space.
4. Our **Route Evaluator** evaluates every candidate route against a multi-objective cost function that balances total distance, travel time, vehicle payload capacity limits, and customer delivery windows.
5. Finally, the optimized routes and turn-by-turn geometry are stored in an **SQLite database** and rendered on our **React 19 Leaflet dashboard**. Dispatchers can also start a live simulation to watch vehicles travel along their routes in real time via WebSockets.

Empirical testing on Pune delivery clusters shows our system achieves **between 1.5% and 14.8% distance reduction** compared to baseline dispatching, while reducing route planning time from nearly an hour to less than two seconds. Our codebase includes **18 automated pytest test cases**, all passing with zero errors.

In the future, we plan to ingest live road traffic sensors and connect physical OBD-II GPS trackers.

Thank you, and I am now ready for your questions."

Chapter 23: Quick Revision Notes (One-Page Cheat Sheet)

- **Project Title:** AI-Based Complexity-Aware Delivery Route Optimization System (`RouteOptimaAI`)
- **Core Problem:** Solving the NP-hard Capacitated Vehicle Routing Problem with Time Windows (CVRPTW).
- **Tech Stack:**
  - **Backend:** Python 3.10+, FastAPI 0.115, Uvicorn, SQLAlchemy 2.0
  - **Frontend:** React 19, Vite 8, Leaflet 1.9, Vanilla CSS (Light Enterprise Design System)
  - **Database:** SQLite (`delivery.db`) with 11 tables (`orders`, `vehicles`, `routes`, etc.)
  - **ML Model:** Random Forest Classifier (`model.joblib`), 85.0% accuracy, 8 features.
- **DAA Algorithms Implemented:**
  1. *Greedy Nearest Neighbor:* Priority-weighted local minimum search ($\mathcal{O}(n^2)$).
  2. *2-Opt Local Search:* Intra-route edge inversion + inter-route customer relocation.
  3. *Genetic Algorithm:* Order Crossover (OX), tournament selection ($k = 3$), elitism ($e = 2$).
  4. *Hybrid:* 80% GA global exploration + 20% 2-Opt local untangling.
- **Objective Cost Function:**

$$\text{Cost} = (1.0 \times \text{Distance}) + (0.2 \times \text{Time}) + (50.0 \times \text{Late Orders}) + (1000.0 \times \text{Capacity Overrun})$$

- **Complexity Score Thresholds:**
  - `0 - 39.9`: **LOW** → Greedy
  - `40.0 - 69.9`: **MEDIUM** → Local Search (2-Opt)
  - `70.0 - 100`: **HIGH** → Genetic / Hybrid
- **Top 3 Advantages:**
  1. Automated address geocoding with local ground truth (no manual coordinates needed).
  2. Dynamic ML matching of algorithm to problem scale saves CPU and improves route quality.
  3. Strict capacity and deadline validation with persistent audit history and live simulation.
- **Top 3 Limitations:**
  1. Static OSRM travel times instead of live real-time traffic jams.
  2. Vehicles return to their own originating depot (no cross-depot handoffs).
  3. Software simulation over WebSockets rather than physical OBD-II hardware GPS trackers.
- **Key Viva Answer to Remember:** *"Dijkstra's algorithm finds the shortest path between two points. It does not sequence multi-stop deliveries. Our DAA algorithms (Greedy, 2-Opt, Genetic) solve the combinatorial sequencing problem using the distance matrix that Dijkstra/OSRM provides."*